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supported by the **hind limbs**, while the bird is at rest or walking; the hind limbs are therefore attached anteriorly from the trunk to balance the body and support the weight of the body at rest. They are warm blooded or **homeothermic**.

Exoskeleton

The exoskeleton of pigeon is derived from the **epidermis** and occurs in the form of **horny claws, scales** and **feathers**. Beaks are used for ingestion, fighting and preening of feathers. Claws are used for walking and perching. Epidermal scales are present on the foot and the entire body is covered by feathers. Arrangement of feathers on the body of bird is called **pterylosis**. Feathers are of three kinds: large **quill feathers** on wings and tail which are used for flight; **contour feathers**, form a covering for the body and **filoplumes,** lie between the contour feathers. The **nestlings** are covered with **down feathers** which resemble the filoplumes.

Structure of a Quill feather

The quill feather has a **stem** or **scapus** and is divided into a lower hollow part called **calamus** or **quill** and an upper solid portion called **rachis**. Lower end of the stem has an opening called **inferior umbilicus** which receives a dermal papilla, supplying nutrients and pigments for the growing feathers (Figure 4.28)

A second opening the **superior umbilicus** occurs at the junction of the quill and the rachis, on the inner face of the feather; close to this opening is a small tuft of soft feathers called **after shaft**. Attached to the rachis are small **filament** or **barbs**; the rachis with the barbs constitute the **vane** or

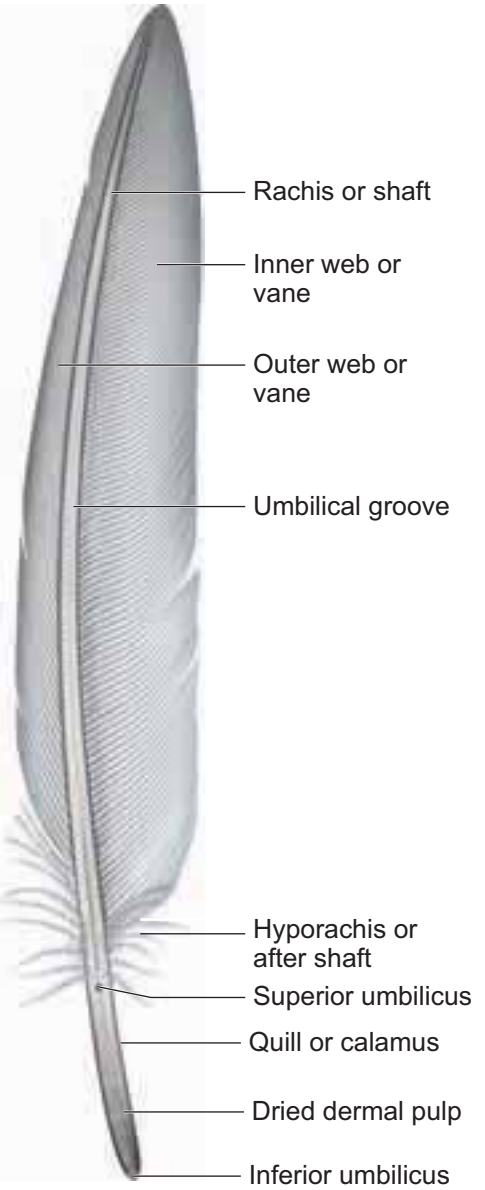

Figure 4.28 Quill Feather

the **vexillum**. Each barb is fringed with an oblique set of processes called **barbules**, which have minute **hooklets** or **barbicels** by which adjacent barbs are hooked together to form a continuous blade for striking the air during flight.

Anatomy

Endoskeleton

The skeletal system is strong but lightly built. The bones are light and spongy. Many of the long bones contain air instead of